WORKSHEET 4.6

Tangent-Line Approximation

Unit 4: Derivatives in Context

Key ideas

- Near $x = a$, $f(x) \approx L(x) = f(a) + f'(a)(x - a)$: the tangent line stands in for the curve.
- Choose a where $f(a)$ and $f'(a)$ are easy and close to the value wanted.
- If the graph is **concave up** near a, the tangent line lies below it: an **underestimate**. Concave down: an **overestimate**.
- Justify over or under with the sign of f'' on the interval, never with the sign of f'.

Worked examples

Example 1 · A square root

Use a tangent line to approximate $\sqrt{26}$.

Solution. $f(x) = \sqrt{x}$ at $a = 25$: $f(25) = 5$, $f'(25) = \dfrac{1}{10}$.

$\sqrt{26} \approx 5 + \dfrac{1}{10}(1) = 5.1$**.**

Example 2 · Over or under

Use the tangent line to $f(x) = \sin x$ at $x = 0$ to approximate $\sin(0.1)$. Is it an overestimate?

Solution. $L(x) = x$, so $\sin(0.1) \approx 0.1$. $f''(x) = -\sin x < 0$ for $0 < x < 0.1$.

0.1**, an overestimate: the graph is concave down there.**

Example 3 · From values

Given $f(2) = 7$ and $f'(2) = -3$, approximate $f(2.2)$.

Solution. $7 + (-3)(0.2)$.

6.4**.**

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Practice

CORE

1. Write the tangent-line approximation $L(x)$ for $f(x) = x^3$ at $x = 2$.

2. Use your answer to Question 1 to approximate $(2.1)^3$.

3. Given $f(5) = 4$ and $f'(5) = 0.5$, approximate $f(4.8)$.

4. A graph is concave up near $x = a$. Is a tangent-line approximation there an overestimate or an underestimate?

EXAM LEVEL

5. Use the tangent line to $f(x) = \dfrac{1}{x}$ at $x = 2$ to approximate $\dfrac{1}{2.1}$. Is it an overestimate or an underestimate? Justify.

6. Use the tangent line to $f(x) = \ln x$ at $x = 1$ to approximate $\ln(0.9)$. Is it an overestimate or an underestimate? Justify.

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7. A function has $f(1) = 2$ and $f'(x) = x^2 + \ln x$ for $x > 0$. Approximate $f(1.1)$ and say whether the approximation is an overestimate or an underestimate.

8. The point $(3, 1)$ lies on the curve $x^2 + y^3 = 10$. Use the tangent line there to approximate the y-coordinate of the point on the curve with $x = 3.1$.

Question 9 · Free-response practice

Let $f(x) = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

(a) Write an equation of the line tangent to the graph of f at $x = \frac{\pi}{4}$.

(b) Use the tangent line to approximate $\tan\left(\frac{\pi}{4} + 0.05\right)$.

(c) Is the approximation in part (b) an overestimate or an underestimate? Justify your answer.

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Worked solutions

1. $L(x) = 8 + 12(x - 2)$.
2. $8 + 12(0.1) = 9.2$.
3. $4 + 0.5(-0.2) = 3.9$.
4. An underestimate.
5. $f(2) = \frac{1}{2}$, $f'(2) = -\frac{1}{4}$: $0.5 - 0.25(0.1) = 0.475$. $f''(x) = \frac{2}{x^3} > 0$, so concave up: an underestimate.
6. $L(x) = x - 1$, so $\ln(0.9) \approx -0.1$. $f''(x) = -\frac{1}{x^2} < 0$, so concave down: an overestimate.
7. $f'(1) = 1$, so $f(1.1) \approx 2.1$. $f''(x) = 2x + \frac{1}{x} > 0$ near 1: an underestimate.
8. $2x + 3y^2\frac{dy}{dx} = 0$ gives $\frac{dy}{dx} = -\frac{6}{3} = -2$ at $(3, 1)$, so $y \approx 1 - 2(0.1) = 0.8$.
9. (a) $f\left(\frac{\pi}{4}\right) = 1$ and $f'\left(\frac{\pi}{4}\right) = \sec^2\frac{\pi}{4} = 2$: $y = 1 + 2\left(x - \frac{\pi}{4}\right)$.
   (b) $1 + 2(0.05) = 1.1$.
   (c) $f''(x) = 2\sec^2 x \tan x > 0$ for x near $\frac{\pi}{4}$, so the graph is concave up and the tangent line lies below it: an underestimate.

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